

Skill Arena - Phase 7 Document 1

Maze Generation, Difficulty Engineering, House Probability & Competitive Challenge Architecture

Document Purpose

Define the core intellectual property responsible for maze generation, difficulty scaling, challenge fairness and house challenge balancing.

Business Objective

Create an infinitely scalable challenge engine that remains fair for players while maintaining long-term platform sustainability.

Core Philosophy

No two challenges should be predictably identical. Every maze must be unique, verifiable and server-generated.

Server Authority

All maze generation, validation, difficulty scoring and completion verification occur server-side.

Maze Generation Formula

Maze Seed + Difficulty Score + Category + Challenge Rules + Randomization Engine = Generated Challenge.

Challenge Metadata

Maze ID, Seed, Difficulty Score, Validation Hash, Category, Replay Reference and Generation Timestamp.

Duplicate Prevention

Generated mazes receive a uniqueness hash. Similarity checks prevent repeated layouts and predictable patterns.

Maze Hash Database

Historical challenge hashes stored permanently to reduce pattern reuse.

Difficulty Engineering

Difficulty is determined by multiple variables rather than maze size alone.

Difficulty Variables

Path Complexity, Decision Density, Dead Ends, False Routes, Trap Density, Visibility Restrictions, Lives and Environmental Complexity.

Infinite Progression

Difficulty operates on an unlimited score model rather than fixed levels.

Difficulty Scaling

Higher categories increase complexity, route ambiguity, trap frequency and solution difficulty.

PvP Challenge Rules

Player versus Player challenges have no timer. Both players receive the same maze. Winner is determined by fastest verified completion.

PvP Fairness Controls

Opponent movement is hidden. Progress indicators may be blurred or delayed to prevent copying.

PvP Completion Logic

Completion times compared after validation. Fastest verified completion wins the challenge.

House Challenge Rules

House challenges become available only after progression requirements are met.

House Challenge Environment

House challenges include timers, lives and enhanced difficulty controls.

AI Solver Framework

Every generated maze is solved by internal solver systems to calculate optimal completion paths.

Human Performance Model

Estimated human completion times derived from solver outputs and historical player performance.

Human Multiplier

Initial configuration uses a 20 percent multiplier above optimal solver time to establish challenge limits.

House Probability Model

Target design objective is approximately 65 percent house win probability and 35 percent player success probability.

House Balance Philosophy

Challenges must remain difficult but achievable. House advantage should come from challenge design, not impossible requirements.

Adaptive Difficulty Monitoring

Completion statistics continuously analyzed to prevent excessive player losses or unintended exploitation.

Replay Verification

Every challenge generates replay data for dispute resolution and integrity verification.

Challenge Integrity

Rewards only issued after completion verification and anti-cheat validation.

Treasury Integration

House challenge payouts validated against treasury and risk management systems.

Monitoring Requirements

Track completion rates, difficulty distribution, house performance and player success rates.

Audit Requirements

All challenge generation, difficulty calculations and outcomes permanently logged.

Acceptance Criteria

Platform can generate unique, scalable and balanced challenges while maintaining long-term fairness and sustainability.